Data Analyst Job Market Analysis – Workshop 3

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Introduction

This report analyses a sample of 400 Data Analyst job postings scraped from Glassdoor (United States, June 2020) to explore how experience requirements and programming-language requirements relate to estimated salaries. Two variables, the minimum years of experience and the languages required (R, Python, Both or Neither), were extracted from the free-text job descriptions with a large language model (Google Gemini 2.5 Flash via OpenRouter) called directly from Google Sheets.

Data and methods

Each job description was sent to the model with a structured prompt through a custom `MYGPT` function in Google Apps Script, and the results were frozen as values. Not every posting produced a usable answer:

Table 1. Extracted variables

Extracted variable	Usable values	Not stated in posting (NA)	Extraction failed	Total
Minimum years of experience	301	81	18	400
Data languages (R / Python / Both / Neither)	246	–	154	400

Failed extractions were caused by exhausted API credits and were excluded from the analysis. The language labels were cross-checked against a simple keyword search; one clear error (a posting mentioning "R&D" classified as using R) was corrected manually to Python. The salary variable is Glassdoor's median salary estimate, available for all 400 postings. Statistics and figures were produced with the help of an AI assistant (Claude) and checked by the author.

Years of experience and salary

In this sample, the minimum experience required is only weakly related to the estimated salary: each additional year is associated with about +$539 on the linear trend, and the rank correlation is close to zero (Spearman $\rho = 0.07$, $p = 0.24$, $n = 301$).

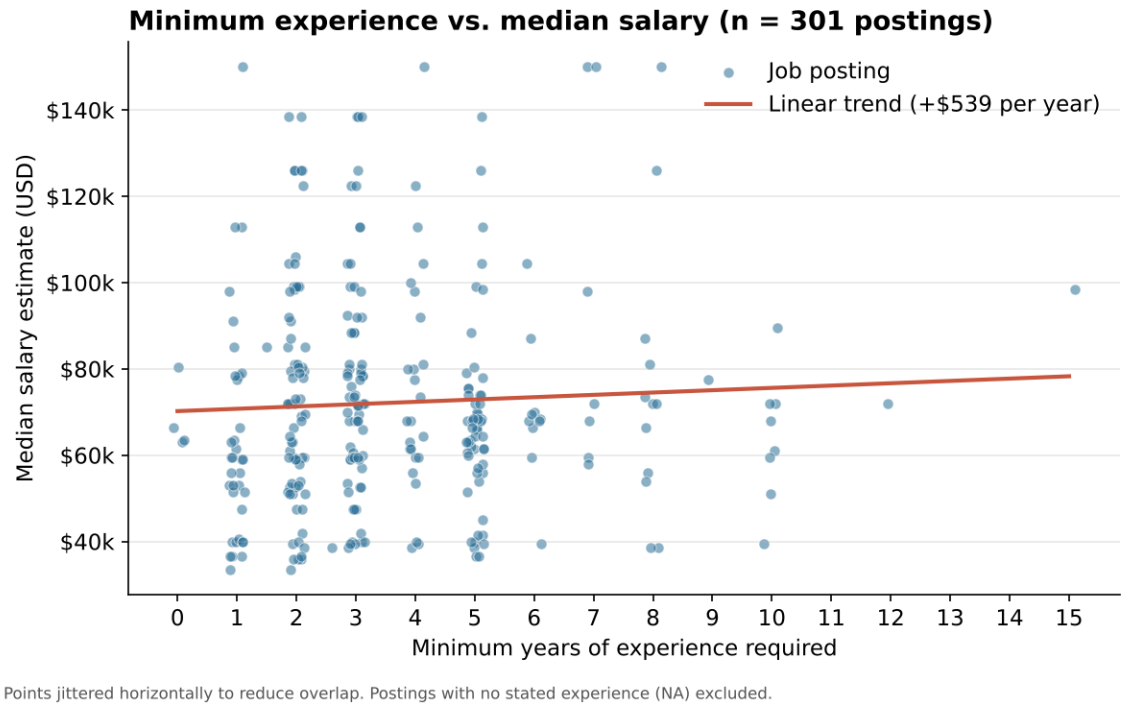


Figure 1. *Minimum years of experience vs. median salary estimate (n = 301 postings; NA and failed extractions excluded).*

Most postings ask for little experience: the median requirement is 3 years (interquartile range 2–5 years), and only 40 postings ask for 6 years or more. Entry-level postings (0–1 year, n = 39) have a lower median salary ($59,500) than postings asking for 2–3 years ($72,000), but beyond that point salaries do not keep rising ($66,500 for 4–5 years, $69,750 for 6+ years). At every experience level, salaries spread widely, from about $35k to $150k.

Interpretation. Experience on its own explains very little of the salary variation in these postings. Other factors probably matter more, such as location, company size, industry and job title seniority. The salary is also a Glassdoor estimate, not the salary actually offered. With only 301 usable postings, and few at the high end of experience, these results are indicative rather than conclusive.

Programming language requirements and salary

Median salaries are similar across the four groups, and the differences are not statistically significant (Kruskal–Wallis H = 1.99, p = 0.57, n = 246).

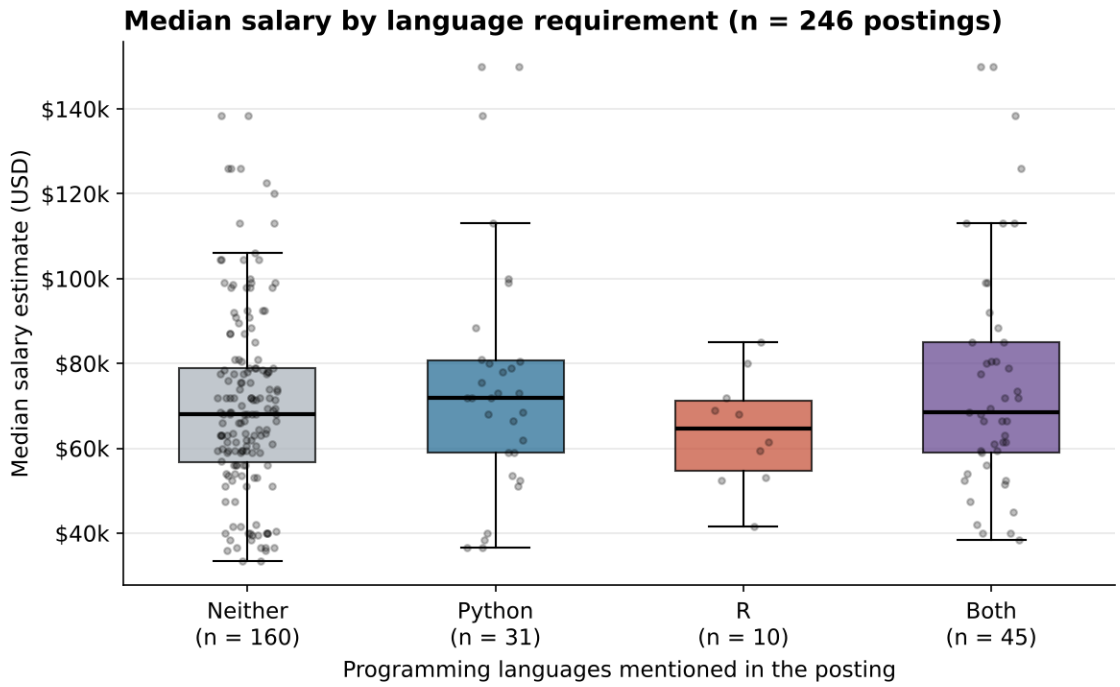


***Figure 2**. Median salary estimate by programming languages mentioned (n = 246 postings; failed extractions excluded). Boxes show the interquartile range, the thick line the median, dots individual postings.*

Two thirds of the classified postings (160 of 246) mention neither R nor Python, which suggests that many Data Analyst roles in 2020 relied mainly on Excel, SQL and BI tools. Postings mentioning Python only (n = 31) have the highest median salary ($72,000), followed by Both ($68,500, n = 45) and Neither ($68,000). Postings mentioning R only are rare (n = 10) and have the lowest median ($64,750). The highest salaries in the sample ($150k) appear only in the Python and Both groups.

Interpretation. Python appears more often than R in these postings and is associated with a slightly higher salary range, but the gaps are small compared with the variation inside each group. The R-only group is far too small (10 postings) to support any conclusion, and 154 postings could not be classified, which may bias the comparison.

Limitations

- **Small and incomplete sample.** 400 postings, of which only 301 (experience) and 246 (languages) could be analysed; subgroups such as R-only (n = 10) or 6+ years (n = 40) are very small.
- **Missing data not at random?** 154 language labels failed because API credits ran out partway through; if the failed rows differ systematically from the others, the comparison is biased.
- **Estimated salaries.** Glassdoor's median salary estimate is a modelled range, not the salary in the offer.

- **Context.** US postings from June 2020 only; the market and the demand for Python and R have changed since.
- **LLM extraction errors.** The model was accurate in spot-checks but not perfect (for example, "R&D" read as R once); not every row was verified by hand.
- **Association, not causation.** These comparisons do not control for location, company size, industry or seniority.

Personal reflection

Overall, the tasks in this workshop were somewhat difficult for me. However, the steps were clear enough to help me understand the process, and whenever a task was unclear, I asked an AI assistant for clarification. What surprised me most was how the model handled contract durations: in one posting, "Duration: 12 months+" was correctly not treated as years of experience, and the model returned NA.

Once the setup was working, extracting the data with the LLM felt simple. I learned a lot from this exercise, and I am confident that I will now be able to repeat these steps easily in my own work.